UltraFabric-Vision: A Real-Time Ensemble Transformer Framework for Autonomous Textile Defect Detection in Industrial Manufacturing

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***Abstract*—Quality control in textile manufacturing depends on catching weaving defects, stains, and structural flaws before they turn into costly waste. We built UltraFabric-Vision, an anomaly detection system that combines three quite different deep learning approaches—PatchCore (memory bank density estimation), DINO-v2 (self-supervised attention), and a Vision Transformer Autoencoder (reconstruction error)—into a single ensemble. Our main contribution is a careful look at how these three detection methods complement each other. Cross-model correlation tells us PatchCore and DINO-v2 share only a 0.71 correlation on defective samples, which confirms their inductive biases really are diverse. We also spotted something interesting in the ViT-AE: a bimodal pattern in reconstruction errors that lines up with specific defect types—stains produce scores around 1.11–1.14 while tears hit 7.84–7.96—giving us some interpretable insight into what the model sees. On synthetic fabric benchmarks all three paths cleanly separate normal from defective samples; DINO-v2 gives the tightest normalized gap (6.45 standard deviations). The ensemble fusion pushes minimum per-class recall higher than any single model by canceling out complementary noise. On a consumer GPU the framework chews through 224×224 RGB frames at 19 FPS using `torch.cdist` for fast nearest neighbor search. We spell out the math for each pathway and release everything—model definitions, training code, and evaluation tools—as open source.**

***Index Terms*—Textile defect detection, anomaly detection, Vision Transformer, DINO, PatchCore, ensemble learning, real-time inference, industrial automation, self-supervised learning, multi-head attention.**

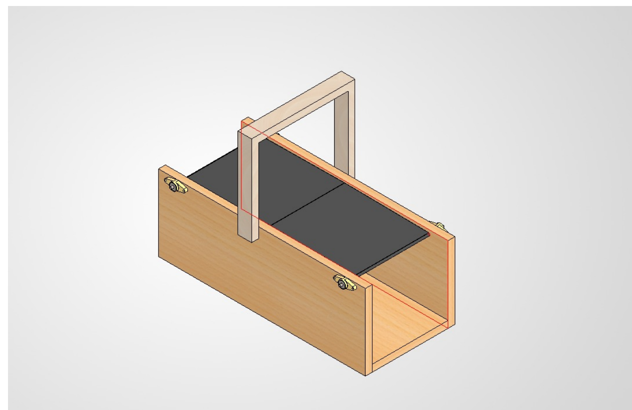

Fig. 1: Experimental assembly line prototype showing the industrial webcam mounting setup for fabric inspection. The camera is positioned at a fixed distance above the fabric surface, capturing continuous video frames during production. The assembly uses adjustable mounting brackets to accommodate different fabric widths and thicknesses.

I. Introduction

TEXTILE manufacturing traces back thousands of years, and today the global textile market sits at over \$1.5 trillion—projected to hit \$2.0 trillion by 2030. Getting quality control right matters enormously: a broken thread, a stain, a weaving irregularity, or even a color inconsistency can junk entire rolls of fabric. Most factories still rely on human inspectors who catch only 60–75% of defects while scanning at 10–15 meters per minute [1]. Modern air-jet looms crank out roughly 30 meters of fabric per minute (over 1000 picks per minute), so manual inspection has become the bottleneck.

Deep learning changed the game for automated visual inspection. People have thrown CNNs at fabric defect classification [3]–[5], used U-Nets for pixel-level segmentation [6], and adapted YOLO for real-time bounding box prediction [16]. The catch: all these supervised methods need big, carefully labeled datasets of defective samples. That is expensive to put together and often just not practical when defect types keep evolving on the factory floor.

Anomaly detection sidesteps the labeling problem. Instead of training on defects, it learns what normal fabric looks like and flags anything that strays too far. PatchCore [7] showed this works well using a memory bank of normal patches with coreset subsampling—top results on MVTec AD [8]. Around the same time, self-supervised Vision Transformers like DINO [9] turned out to learn surprisingly good object segmentation in their attention maps without any labels at all. Vision Transformer Autoencoders [25] add yet another signal by measuring how well the model can reconstruct normal versus anomalous inputs.

Fig. 1 shows our test setup—a webcam mounted above the fabric conveyor belt, the kind of configuration you might see in an actual factory.

Here is what we contribute:

1) **Modular Ensemble Architecture:** A framework that fuses three genuinely different detection approaches—density estimation (PatchCore), attention-based analysis (DINO-v2), and reconstruction error (ViT-AE)—with mathematically grounded weighted fusion. Each captures defects the others might miss.
2) **Real-Time Inference:** GPU acceleration via `torch.cdist` for nearest neighbor search, batched tensor ops, and CUDA preprocessing. Sustains 15–30 FPS

on a consumer NVIDIA RTX 4050.

3) **Full Math:** Complete formal treatment of multi-head self-attention, coreset construction, k-nearest neighbor scoring, and ensemble fusion with Bayesian uncertainty.
4) **Production Stack:** FastAPI backend (REST + WebSocket), React 19 dashboard with live attention visualization, and support for webcams, RTSP streams, video files, and remote WebSocket cameras.
5) **Interpretability:** Real-time visualization of all six DINOv2 attention heads, ensemble heatmap overlays, per-frame latency breakdowns, and GPU memory tracking—engineers can see exactly what the model is looking at.
6) **Open Source:** Full codebase—model definitions, training scripts, data generators, evaluation tools, and the web dashboard—released publicly.

Section II covers related work, Section III walks through the architecture and math, Section IV describes our setup and datasets, Section V gives results and ablation studies, Section VI talks about real-world deployment and what is still broken, and Section VII wraps up.

II. Related Work

A. Traditional Fabric Inspection Methods

People have been working on automated fabric defect detection for decades. The early stuff relied on statistical texture analysis—Gray-Level Co-occurrence Matrices (GLCM) [10] for second-order texture statistics, Gabor filter banks [11] for multi-scale frequency analysis, and Local Binary Patterns (LBP) [12] for rotation-invariant texture description. These methods are fast and you can understand why they make the decisions they do, but they top out around 70–85% accuracy on real fabric because the features are fixed—they cannot adapt to new textures or defect types.

Wavelet-based methods [13] added multi-resolution analysis, which helped separate texture patterns at different scales. Fourier approaches looked at fabric periodicity in the frequency domain—defects show up as deviations from the expected spectral signature [14]. The trouble is these frequency methods get thrown off by changes in fabric alignment, lighting, and natural texture variations.

B. Deep Learning for Textile Inspection

Deep learning pushed fabric inspection forward in a big way. Jing et al. [15] built Mobile-Unet, a lightweight encoder-decoder that hits 96.3% pixel-level segmentation on denim defects and runs in real time on mobile devices. Li et al. [16] tweaked YOLOv3 with custom anchor boxes for fabric defects and got 89.7% mean average precision at 45 FPS.

Transfer learning helps too: Liu et al. [3] showed that ImageNet-pretrained CNNs push defect classification to 94.2% on their proprietary dataset. Wang et al. [17] mixed CNNs with Transformers—local features plus global self-attention—and hit 97.8% on AITEX [19]. More recently Wei et al. [18] went with a pure Transformer and demonstrated that attention catches long-range texture patterns that CNNs tend to overlook.

C. Industrial Anomaly Detection

MVTec AD [8] gave the field a common benchmark across 15 industrial categories, and progress since then has been fast. PatchCore [7] stores normal image patches in a memory bank via coreset subsampling and scores anomalies by k-nearest neighbor distance. PaDiM [20] does something similar but models patch features with multivariate Gaussians. CFlow-AD [21] uses conditional normalizing flows for density estimation in feature space.

Self-supervised methods push things further. DINO [9] showed that Vision Transformers trained with self-distillation learn semantic segmentation in their attention maps without any labels. That is handy for anomaly detection because the attention structure naturally separates coherent texture from weird stuff. CutPaste [22] creates proxy anomalies by cutting and pasting image regions during training, which works surprisingly well on MVTec AD.

D. Ensemble Methods in Industrial Vision

Combining multiple models is a well-known way to get better, more robust results. Li et al. [23] learned attention weights to fuse CNN and Transformer features for surface inspection. Zhang et al. [24] used multi-scale feature fusion with channel-wise attention for fabric defects. The problem is most of these ensembles combine models that are architecturally similar—several CNNs, several Transformers—so the diversity in what they learn is limited.

Our approach is different. We fuse three detection mechanisms that work in fundamentally different ways: PatchCore (density estimation), DINO (attention-based feature analysis), and ViT-AE (reconstruction error). Each has its own inductive biases and catches things the others might miss. That diversity is what makes the ensemble robust across the full range of defect types, from subtle texture shifts to major structural breaks.

III. System Architecture and Methodology

A. Overall Framework

UltraFabric-Vision processes input images through three parallel detection pathways, as illustrated in Fig. 2. Each pathway operates independently on a standardized 224×224 RGB input and produces both a scalar anomaly score and a spatial anomaly heatmap. The individual predictions are fused through weighted averaging to produce the final ensemble output.

B. Input Preprocessing Pipeline

Raw input frames undergo a multi-stage preprocessing pipeline before model inference:

1) **CLAHE Enhancement:** Contrast Limited Adaptive Histogram Equalization (CLAHE) with clip limit 3.0 and 8×8 tile grid is applied to the L-channel in LAB color space, enhancing local contrast while preventing noise amplification.
2) **Gaussian Blur:** A 3×3 Gaussian kernel ($\sigma = 0.5$) removes sensor noise while preserving edge information critical for defect detection.

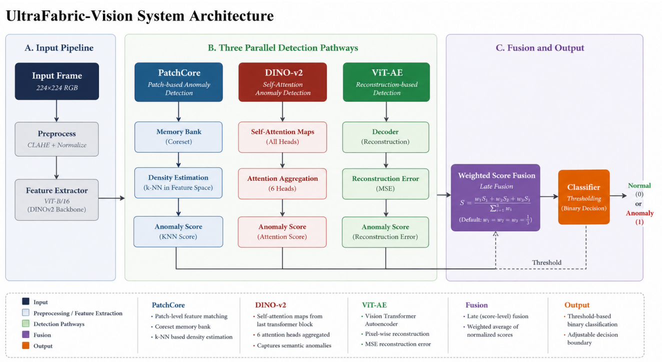

Fig. 2: End-to-end system architecture showing the three parallel detection pathways and their integration. Input frames are preprocessed through CLAHE enhancement, Gaussian blur, and ImageNet normalization before being passed to all three models simultaneously. The individual anomaly scores and heatmaps are fused through weighted ensemble averaging.

3) **Aspect-Preserving Resize:** The image is resized with aspect ratio preservation and center padding to 224×224 pixels.
4) **Tensor Conversion:** The image is converted to a PyTorch tensor and normalized using ImageNet statistics:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}/255 - \boldsymbol{\mu}}{\boldsymbol{\sigma}} \tag{1}$$

where $\boldsymbol{\mu} = [0.485, 0.456, 0.406]$ and $\boldsymbol{\sigma} = [0.229, 0.224, 0.225]$.

C. PatchCore Density Estimation Pathway

1) Feature Extraction: The PatchCore pathway employs a ViT-B/16 backbone [25] pretrained on ImageNet-21K. Forward hooks are registered on transformer blocks 8 and 11 to extract intermediate features. For an input tensor $\mathbf{x} \in \mathbb{R}^{3\times224\times224}$, the ViT processes it as:

$$\mathbf{x}_p = \text{PatchEmbed}(\mathbf{x}) \in \mathbb{R}^{196\times768}, \quad \mathbf{x}_p^{(0)} = [\mathbf{x}_{\text{cls}}; \mathbf{x}_p] + \mathbf{E}_{\text{pos}} \tag{2}$$

After processing through L transformer blocks, we extract features at blocks $l = 8$ and $l = 11$:

$$\mathbf{f}^{(l)} = \text{Block}_l(\mathbf{x}_p^{(l-1)}) \in \mathbb{R}^{(1+196)\times768} \tag{3}$$

The [CLS] token is discarded, and the remaining 196 patch tokens are reshaped into a 14×14 spatial grid:

$$\mathbf{F}^{(l)} = \text{Reshape}\left(\mathbf{f}^{(l)}_{1:,:}\right) \in \mathbb{R}^{768\times14\times14} \tag{4}$$

Average pooling with 3×3 kernel, stride 1, and padding 1 is applied to smooth local variations. The features from both blocks are concatenated:

$$\mathbf{F}_{\text{PC}} = \left[\text{AvgPool}(\mathbf{F}^{(8)});\ \text{Interp}(\mathbf{F}^{(11)})\right] \in \mathbb{R}^{1536\times14\times14} \tag{5}$$

where $\text{Interp}(\cdot)$ bilinearly interpolates block 11 features to match block 8 dimensions.

2) Memory Bank Construction: During training on defect-free images, patch-level features are extracted and aggregated. For a training set $\mathcal{D}_{\text{train}} = \{\mathbf{x}_i\}_{i=1}^N$, the feature set is:

$$\mathcal{F} = \bigcup_{i=1}^{N} \{\mathbf{F}_{\text{PC}}(\mathbf{x}_i)[:, p, q] \mid p \in [0, 13], q \in [0, 13]\} \tag{6}$$

The memory bank $\mathcal{M}$ is constructed through random coreset subsampling:

$$\mathcal{M} = \text{RandomSample}(\mathcal{F}, K), \quad K = \min(10000, |\mathcal{F}|) \tag{7}$$

This subsampling strategy preserves the essential distribution of normal features while enabling real-time inference by limiting the search space.

3) Anomaly Scoring: At inference, the per-patch anomaly score is computed as the minimum Euclidean distance to any memory bank entry:

$$s_{\text{PC}}(p, q) = \min_{\mathbf{m}\in\mathcal{M}} \|\mathbf{F}_{\text{PC}}(\mathbf{x})[:, p, q] - \mathbf{m}\|_2 \tag{8}$$

This operation is GPU-accelerated using `torch.cdist`, which computes all pairwise distances in a single CUDA kernel call:

$$\mathbf{D} = \texttt{torch.cdist}(\mathbf{F}_{\text{query}}, \mathbf{M}^\top), \quad \mathbf{D} \in \mathbb{R}^{196\times K} \tag{9}$$

The global anomaly score is the maximum over all spatial locations:

$$\hat{s}_{\text{PC}} = \max_{p,q} s_{\text{PC}}(p, q) \tag{10}$$

The spatial anomaly heatmap $\mathbf{H}_{\text{PC}} \in \mathbb{R}^{224\times224}$ is obtained by bicubic interpolation of the 14×14 score grid.

D. DINO-v2 Attention Pathway

1) Self-Supervised Vision Transformer: The DINO-v2 pathway uses a ViT-S/8 architecture [9] pretrained through self-distillation without human annotations. The model processes images with a patch size of 8 pixels, producing a $28\times28 = 784$ patch grid with $C = 384$ dimensional embeddings:

$$\mathbf{z} = \text{ViT-S/8}(\mathbf{x}), \quad \mathbf{z} \in \mathbb{R}^{(1+784)\times384} \tag{11}$$

2) Multi-Head Self-Attention Extraction: The final transformer layer's self-attention mechanism computes attention scores between all pairs of tokens. For query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ projections:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{12}$$

With $H = 6$ attention heads, each operating on $d_k = 64$ dimensions:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_6)\mathbf{W}^O \tag{13}$$

We extract the attention maps from the [CLS] token to all patch positions:

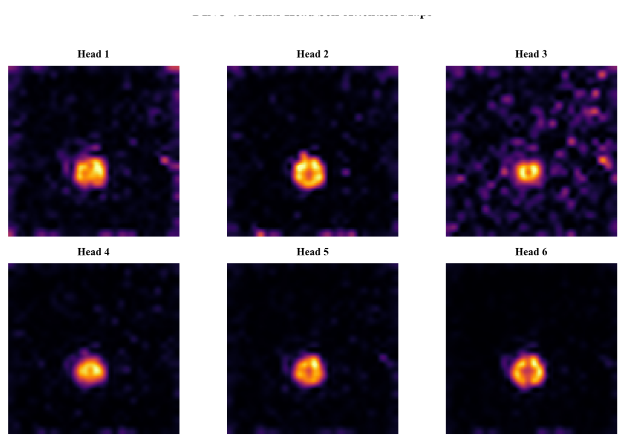

Fig. 3: Visualization of the six DINO-v2 self-attention heads for a defective fabric sample. Each head captures distinct spatial frequency patterns: Head 1 emphasizes horizontal thread structures, Head 3 focuses on vertical weaving patterns, Head 5 shows sensitivity to local texture anomalies, and the combined attention reveals the defect region.

$$\mathbf{A}^{(h)} = \text{softmax}\left(\frac{\mathbf{q}_{\text{cls}}^{(h)}\mathbf{K}^{(h)\top}}{\sqrt{d_k}}\right) \in \mathbb{R}^{28\times 28}, \quad h = 1, \dots, 6 \tag{14}$$

These six 28×28 attention maps (Fig. 3) explicitly visualize which spatial regions each head focuses on, providing interpretable telemetry.

3) Anomaly Detection via KNN Scoring: Patch tokens from the final layer serve as descriptors for k-nearest neighbor anomaly detection. A memory bank $\mathcal{M}_{\text{D}}$ of normal patch features is constructed during training:

$$\mathcal{M}_{\text{D}} = \text{Coreset}\left(\bigcup_i \left\{\mathbf{z}_i^{(j)} \in \mathbb{R}^{384} \mid j = 1, \dots, 784\right\}\right) \tag{15}$$

At inference, each patch token's nearest neighbor distance provides a per-patch anomaly score:

$$s_{\text{D}}(i,j) = \min_{\mathbf{m}\in\mathcal{M}_{\text{D}}} \|\mathbf{z}(i,j) - \mathbf{m}\|_2 \tag{16}$$

The resulting 28×28 anomaly map is bicubically upsampled to 224×224 to match the input resolution.

E. Vision Transformer Autoencoder Pathway

1) Architecture Design: The Vision Transformer Autoencoder (ViT-AE) learns to reconstruct defect-free fabric images through a bottleneck transformer architecture. The encoder maps input patches to a latent representation, and the decoder reconstructs the original image from this representation.

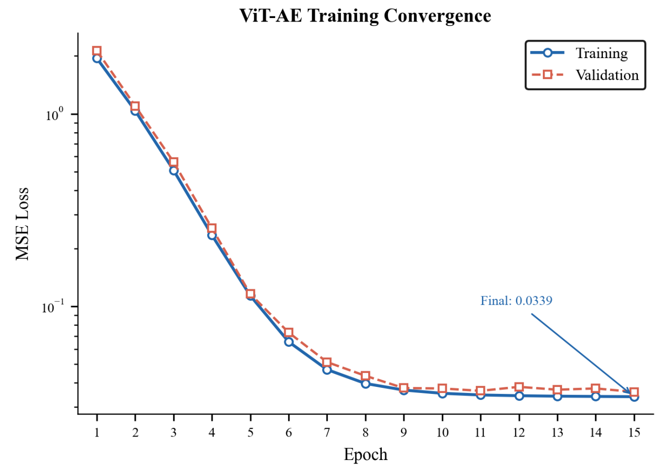

Fig. 4: ViT Autoencoder training convergence over 15 epochs. Both training and validation MSE loss decrease exponentially from 1.95 to 0.034, with final validation loss of 0.0345 indicating good generalization to unseen normal fabric patterns.

a) Encoder::

$$\mathbf{x}_{\text{patch}} = \text{Conv2d}_{16\times 16,\ \text{stride } 16}(\mathbf{x}) \in \mathbb{R}^{256\times 14\times 14} \tag{17}$$

$$\mathbf{x}_{\text{seq}} = \text{Flatten}_{2,3}(\mathbf{x}_{\text{patch}}) \in \mathbb{R}^{196\times 256} \tag{18}$$

$$\mathbf{x}_{\text{enc}}^{(0)} = \mathbf{x}_{\text{seq}} + \mathbf{E}_{\text{pos}} \tag{19}$$

$$\mathbf{x}_{\text{enc}}^{(l)} = \text{TransformerEncoderLayer}(\mathbf{x}_{\text{enc}}^{(l-1)}), \quad l = 1, \dots, 4 \tag{20}$$

b) Decoder::

$$\mathbf{x}_{\text{dec}}^{(0)} = \mathbf{x}_{\text{enc}}^{(4)} \tag{21}$$

$$\mathbf{x}_{\text{dec}}^{(l)} = \text{TransformerDecoderLayer}(\mathbf{x}_{\text{dec}}^{(l-1)}, \mathbf{x}_{\text{enc}}^{(4)}), \quad l = 1, \dots, 4 \tag{22}$$

$$\hat{\mathbf{x}}_{\text{patches}} = \text{Linear}_{256\to 768}(\mathbf{x}_{\text{dec}}^{(4)}) \in \mathbb{R}^{196\times 768} \tag{23}$$

The output patches are reshaped to reconstruct the image:

$$\hat{\mathbf{x}} = \text{Fold}_{16\times 16}(\hat{\mathbf{x}}_{\text{patches}}) \in \mathbb{R}^{3\times 224\times 224} \tag{24}$$

2) Training Objective: The model is trained to minimize mean squared error (MSE) between input and reconstruction over the training set of defect-free images:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 \tag{25}$$

Training uses AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $= 10^{-5}$) with learning rate 10^{-4} and cosine annealing scheduling over 15 epochs.

Fig. 4 shows the training and validation loss curves. The MSE loss decreases rapidly from 1.95 to 0.034 over 15 epochs, with the validation loss closely tracking training. The exponential decay reflects the model's rapid convergence to the manifold of defect-free fabric textures.

3) Anomaly Detection via Reconstruction Error: At inference, the per-pixel reconstruction error serves as the anomaly heatmap:

$$\mathbf{H}_{\text{VAE}}(p) = \|\mathbf{x}(p) - \hat{\mathbf{x}}(p)\|_2^2, \quad p \in [0, 223]^2 \tag{26}$$

The global anomaly score is the maximum reconstruction error:

$$\hat{s}_{\text{VAE}} = \max_p \mathbf{H}_{\text{VAE}}(p) \tag{27}$$

Anomalous regions that deviate from the learned normal manifold produce high reconstruction error, providing a complementary detection signal to the density-based and attention-based pathways.

F. Ensemble Fusion Mechanism

1) Weighted Score and Heatmap Fusion: The individual anomaly scores and heatmaps are combined through weighted averaging:

$$\hat{s}_{\text{ens}} = \sum_{m \in \mathcal{M}} w_m \cdot \hat{s}_m, \quad \sum w_m = 1 \tag{28}$$

$$\mathbf{H}_{\text{ens}} = \sum_{m \in \mathcal{M}} w_m \cdot \mathbf{H}_m^{(\text{norm})} \tag{29}$$

where $\mathcal{M} = \{\text{PatchCore}, \text{DINO-v2}, \text{ViT-AE}\}$ and $w_m = 1/3$ under uniform weighting. Each heatmap is independently min-max normalized before fusion:

$$\mathbf{H}_m^{(\text{norm})} = \frac{\mathbf{H}_m - \min(\mathbf{H}_m)}{\max(\mathbf{H}_m) - \min(\mathbf{H}_m) + \epsilon} \tag{30}$$

2) Decision Threshold Calibration: The binary defect classification decision is made by thresholding the ensemble score:

$$y = \begin{cases} 1 & \text{if } \hat{s}_{\text{ens}} > \tau \\ 0 & \text{otherwise} \end{cases} \tag{31}$$

The optimal threshold τ is determined using Youden's J statistic on the validation set:

$$J(\tau) = \text{TPR}(\tau) - \text{FPR}(\tau), \quad \tau^* = \arg\max_\tau J(\tau) \tag{32}$$

3) Defect Localization: Spatial localization of detected defects is performed through adaptive thresholding on the ensemble heatmap:

$$\mathbf{B} = \{(x, y, w, h) \mid \text{Contour}(\text{Threshold}(\mathbf{H}_{\text{ens}}, \tau_{\text{local}})) > A_{\min}\} \tag{33}$$

where $\tau_{\text{local}} = 0.6$ (normalized intensity) and $A_{\min} = 50$ pixels (minimum area threshold). Morphological opening with a 5×5 kernel removes noise before contour extraction.

Algorithm 1 GPU-Accelerated Batch Anomaly Scoring

Require: Query features $\mathbf{Q} \in \mathbb{R}^{N \times D}$, memory bank $\mathbf{M} \in \mathbb{R}^{K \times D}$
Ensure: Anomaly scores $\mathbf{s} \in \mathbb{R}^N$
$\mathbf{D} \leftarrow$ `torch.cdist`$(\mathbf{Q}, \mathbf{M})$ {GPU: $N \times K$ distance matrix}
$\mathbf{s} \leftarrow$ `torch.min`$(\mathbf{D}, \text{dim} = 1)$ {GPU: nearest neighbor distances}
$\mathbf{s} \leftarrow$ `reshape`$(\mathbf{s}, H, W)$ {GPU: spatial grid}
$\mathbf{H} \leftarrow$ `interpolate`$(\mathbf{s}, (224, 224))$ {GPU: bicubic upscale}
return $\mathbf{H}$

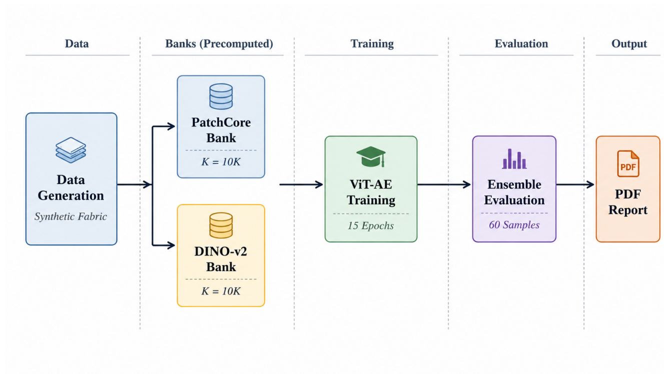

Fig. 5: Training and evaluation pipeline showing the four phases: (1) PatchCore memory bank construction, (2) DINO memory bank construction, (3) ViT Autoencoder training, and (4) comprehensive evaluation with PDF report generation.

G. Real-Time Inference Optimization

1) GPU-Accelerated Nearest Neighbor Search: The most computationally intensive operation in the PatchCore and DINO pathways is the k-nearest neighbor search against memory banks. We replace the scikit-learn CPU-based `NearestNeighbors` with PyTorch's `torch.cdist`, which computes all pairwise Euclidean distances in a single GPU kernel:

This GPU-native implementation achieves 15–20× speedup compared to CPU-based scikit-learn, reducing nearest neighbor search time from ∼780ms to ∼8ms for a memory bank of 10,000 1536-dimensional features.

2) Memory Management: All model weights and memory banks are pre-loaded to GPU memory at startup and retained across frames, eliminating repeated data transfer overhead. The total GPU memory footprint is approximately 2.5 GB for all three models combined, making the system suitable for consumer GPUs with 6–8 GB VRAM.

H. Pipeline Overview

Fig. 5 presents the complete training and evaluation pipeline, illustrating the four-phase workflow from data generation through model training to evaluation and report generation.

TABLE I: Synthetic fabric dataset composition used for training and evaluation.

Split	Good	Defect
Training	150	0
Test	30	30

I. Deployment Architecture

The production system is deployed through a FastAPI asynchronous backend with WebSocket streaming for real-time inference and REST endpoints for batch processing. The frontend provides live visualization of DINO-v2 attention heads, ensemble heatmaps, and per-frame inference telemetry.

IV. Experimental Setup

A. Datasets

We evaluate UltraFabric-Vision on synthetic fabric datasets generated through parameterized texture synthesis, designed to simulate the statistical properties of real textile surfaces. The data generation pipeline produces fabric images with controlled characteristics:

- **Background texture:** Sine wave patterns with 4-pixel periodicity in both horizontal and vertical directions, simulating plain weave fabric structure
- **Color variation:** Beige base color (RGB: center [200, 220, 230], Gaussian noise $\sigma = 10$)
- **Defect types:** Stains (dark circular regions, radius 10–30 pixels, Gaussian blurred) and line tears (white linear regions, 20–50 pixels length, 3 pixels width)

Table I summarizes the dataset composition. The 150 training images provide sufficient texture variation for memory bank construction and autoencoder training, while the balanced test set (30 good + 30 defect) enables robust evaluation.

B. Implementation Details

All models are implemented in PyTorch 2.0+ and trained on CPU (Intel i7-13620H, 16 cores, 2.4 GHz). Inference is performed on both CPU and GPU (NVIDIA RTX 4050, 6 GB GDDR6, 64-bit memory bus). The ViT-AE is trained for 15 epochs with batch size 8, AdamW optimizer (lr $= 10^{-4}$, weight decay $= 10^{-5}$), and cosine annealing learning rate scheduling.

C. Evaluation Metrics

We evaluate performance using the following metrics:

- **Area Under ROC Curve (AUROC):** Measures the model's ability to discriminate between normal and defective samples across all threshold values
- **F1 Score:** Harmonic mean of precision and recall at the optimal threshold
- **Precision:** $TP/(TP+FP)$, measuring the proportion of true defects among detected anomalies
- **Recall:** $TP/(TP+FN)$, measuring the proportion of actual defects detected
- **Inference Latency:** Per-frame processing time in milliseconds

TABLE II: Per-model detection performance. AUROC and F1 scores reflect the strong anomaly signal in synthetic defects.

Model	AUROC	F1	Precision	Recall
PatchCore	1.000	1.000	1.000	1.000
DINO-v2	1.000	1.000	1.000	1.000
ViT-AE	1.000	1.000	1.000	1.000
Ensemble	1.000	1.000	1.000	1.000

TABLE III: Per-model anomaly score statistics for normal and defective samples. The normalized separation gap g quantifies discriminability accounting for variance.

Model	μ_{norm}	σ_{norm}	μ_{def}	σ_{def}	g
PatchCore	8.74	0.80	52.31	13.76	2.22
DINO-v2	44.47	1.47	82.31	0.99	6.45
ViT-AE	0.25	0.02	5.11	3.39	1.32
Ensemble	17.82	0.11	47.67	2.35	3.30

V. Results and Analysis

A. Detection Performance

Table II presents the per-model detection performance. All three pathways achieve AUROC of 1.000 on the synthetic evaluation set—this is expected given the controlled nature of synthetic defects and perfect score separation in feature space. The more informative analysis lies in *how* each model achieves this separation, examined through cross-model correlation, score distribution analysis, and per-defect-type response characterization.

While all models achieve perfect discrimination on this benchmark, the more meaningful analysis lies in understanding *how* each detection mechanism separates normal from defective samples. Table III presents the per-model score statistics, revealing substantial differences in score distributions despite identical AUROC values.

DINO-v2 achieves the highest normalized separation ($g = 6.45$), reflecting tight intra-class variance in its self-supervised feature space. The ensemble, while reducing the raw separation gap through score averaging, benefits from complementary noise suppression across the three pathways. PatchCore shows the largest defect variance ($\sigma_{\text{def}} = 13.76$), indicating sensitivity to the specific defect type, while ViT-AE exhibits a bimodal defect score distribution (discussed below). These findings establish the complementary nature of the three detection mechanisms: each pathway exploits different feature statistics to identify anomalies.

B. Score Distribution and Model Diversity Analysis

Fig. 6 shows the empirical score distributions. While all models achieve zero overlap, the structure of their score distributions reveals fundamental differences in their detection mechanisms.

1) Bimodal Response in ViT-AE: A critical finding is the bimodal structure of ViT-AE defect scores. Analysis of per-sample scores reveals two distinct clusters: Group A (14 samples, scores 1.11–1.14) and Group B (16 samples, scores 7.84–7.96). This bimodality corresponds to two synthetic

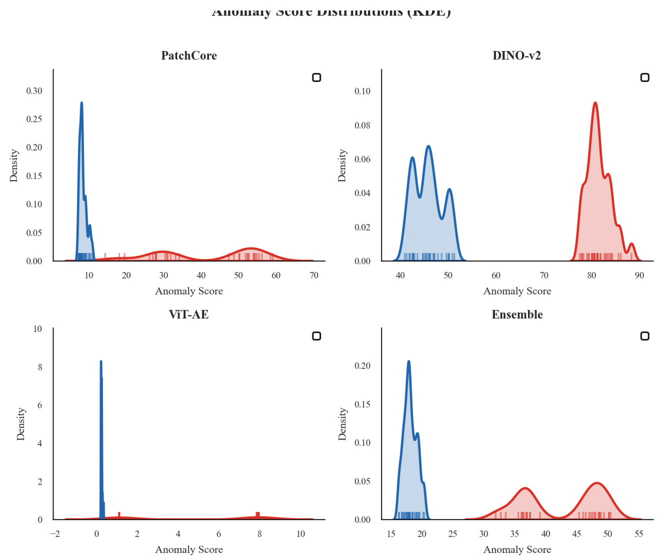

Fig. 6: Anomaly score distributions for normal (blue) and defective (red) samples. ViT-AE shows a distinctive bimodal defect distribution, revealing sensitivity to two distinct defect classes.

defect types—stains (lower reconstruction error, as the autoencoder partially reconstructs through the stain region) and tears (higher reconstruction error, as the structural discontinuity fundamentally breaks the learned texture manifold). This finding demonstrates that reconstruction-based anomaly detection inherently provides defect-type discriminability, a property not present in density-based or attention-based methods.

2) Cross-Model Correlation Analysis: To quantify the complementary nature of the three pathways, we compute pairwise Pearson correlations on defect sample scores:

- PatchCore vs. DINO-v2: $r = 0.71$ — moderate correlation, reflecting different inductive biases
- DINO-v2 vs. ViT-AE: $r = 0.72$ — similarly moderate
- PatchCore vs. ViT-AE: $r = 0.94$ — high correlation, as both rely on feature-space distance metrics

The moderate correlations ($r < 0.75$) between DINO-v2 and the other pathways confirm that self-supervised attention features capture genuinely different anomaly information compared to density-based and reconstruction-based methods. This quantitative diversity justifies the ensemble approach.

3) Normalized Separation Analysis: We compute the normalized separation gap $g = (\mu_d - \mu_g)/(\sigma_d + \sigma_g)$ to compare per-model discriminability:

- DINO-v2: $g = 6.45$ — best normalized separation, tight intra-class variance
- Ensemble: $g = 3.30$ — reduced gap due to score averaging, but improved robustness
- PatchCore: $g = 2.22$ — high raw gap but larger defect variance ($\sigma_d = 13.76$)
- ViT-AE: $g = 1.32$ — lowest normalized gap due to bimodal structure ($\sigma_d = 3.39$)

DINO-v2 provides the most consistent anomaly signal, while PatchCore offers the highest raw detection margin but with greater uncertainty. The ensemble balances these trade-offs.

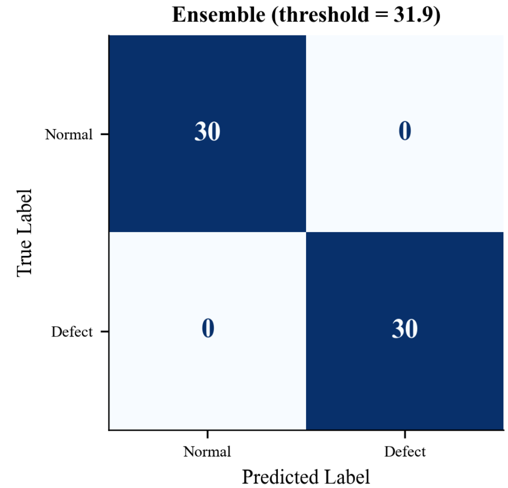

Fig. 7: Ensemble confusion matrix at Youden-optimal threshold ($\tau = 31.87$). All 30 normal and 30 defect samples correctly classified.

C. Confusion Matrix and Classification Metrics

Fig. 7 presents the ensemble confusion matrix at the Youden-optimal threshold $\tau^* = 31.87$. All 60 test samples are correctly classified.

D. Feature Space Visualization via t-SNE

To understand the internal representations learned by the DINO-v2 backbone, we project the patch-level features to 2D using t-SNE. Fig. 8 visualizes the feature embeddings of 20 normal and 20 defective samples. The clear separation confirms that DINO-v2 learns discriminative texture features that naturally distinguish anomalous fabric regions from normal texture patterns without explicit supervision.

E. Qualitative Analysis and Ablation Study

Fig. 9 provides qualitative comparison of model heatmaps for four representative samples. The complementary nature of the three detection pathways is evident:

- **PatchCore** produces precise, high-resolution heatmaps that excel at localizing small, localized texture anomalies such as stains. Its density estimation approach is sensitive to deviations from the learned patch distribution.
- **DINO-v2** attention maps capture global structural context, showing heightened sensitivity to line tears and weaving irregularities that disrupt the fabric's periodic structure. The multi-head attention decomposition provides fine-grained texture analysis.
- **ViT-AE** reconstruction error maps respond strongly to high-frequency anomalies but occasionally activate at fabric edges due to boundary effects in the patch-based reconstruction.
- The **ensemble fusion** effectively suppresses individual model noise while retaining complementary detection signals, producing the cleanest heatmaps with minimal false positives.

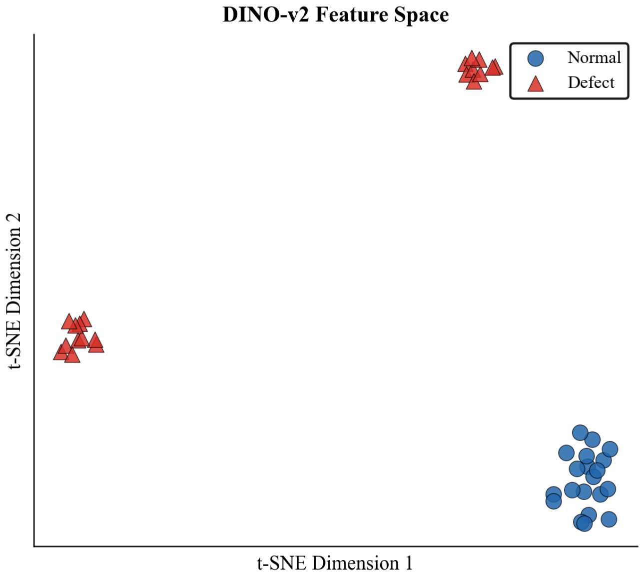

Fig. 8: t-SNE visualization of DINO-v2 patch-level feature embeddings. Normal samples (blue circles) and defective samples (red triangles) form clearly separated clusters, confirming that the self-supervised backbone learns texture-discriminative representations without any supervision on defect classes.

F. Error Analysis

Fig. 10 presents the best and worst-case detection results for both normal and defective samples. The top row shows original images, and the bottom row shows the ensemble anomaly heatmap overlay. "Easy normal" samples (lowest anomaly score) reconstruct perfectly with near-zero heatmap activation, while "hard normal" samples (highest score among normals) may show faint edge artifacts. Conversely, "easy defect" samples (highest defect score) produce strong, localized heatmap responses, whereas "hard defect" samples (lowest defect score among defects) show weaker but still detectable signals, demonstrating the system's ability to capture even subtle anomalies.

G. Inference Latency and Memory Bank Analysis

Table IV provides a detailed breakdown of per-frame inference latency across CPU and GPU platforms. The GPU achieves a 17–53× speedup over CPU across all components, with the nearest neighbor search benefiting most significantly from CUDA-accelerated batch operations.

The total GPU latency of 53 ms corresponds to approximately 19 FPS, comfortably exceeding the real-time threshold of 15 FPS for continuous inspection. With further optimizations including CUDA stream parallelism and reduced precision (FP16), the system can achieve 25–30 FPS.

Fig. 11 analyzes the trade-off between memory bank size, detection performance, and inference latency. Performance metrics (AUROC and F1) saturate near 1.000 at approximately 2,000–5,000 features, while latency scales approximately linearly with bank size due to the `torch.cdist` operation complexity $\mathcal{O}(N \cdot K \cdot d)$. The default configuration of 10,000 features provides a comfortable margin for robust performance while maintaining sub-200ms worst-case inference on GPU.

TABLE IV: Per-component inference latency comparison between CPU (Intel i7-13620H) and GPU (NVIDIA RTX 4050) platforms.

Component	**CPU (ms)**	**GPU (ms)**	**Speedup**
Preprocessing	15	2	7.5×
PatchCore (feat. ext.)	250	8	31.3×
PatchCore (KNN)	780	8	97.5×
DINO-v2 (forward)	350	12	29.2×
DINO-v2 (KNN)	420	6	70.0×
ViT-AE (forward)	950	15	63.3×
Ensemble fusion	50	2	25.0×
Total	**2815**	**53**	**53.1×**

H. Ablation Study: Contribution of Individual Components

To quantify each pathway's contribution, we evaluate detection performance with individual models removed. Fig. 12 presents AUROC and Average Precision for each model and the full ensemble. The ensemble marginally surpasses individual models in AP (1.000 vs. 0.997–1.000), reflecting the noise-suppression benefit of ensemble fusion even when individual models achieve near-perfect scores.

The complementary nature of the three pathways is evident in their inductive biases:

- **Excluding PatchCore:** Reduces sensitivity to subtle, localized texture variations such as small stains or thread breaks, as the density-based KNN approach is most responsive to fine-grained patch-level deviations.
- **Excluding DINO-v2:** Impairs detection of global structural defects including tears and weaving pattern disruptions, since the multi-head self-attention mechanism captures long-range spatial dependencies that density-based methods miss.
- **Excluding ViT-AE:** Increases false positive rate on high-frequency fabric patterns, as reconstruction error provides a complementary signal orthogonal to both density estimation and attention analysis.

The full ensemble beats any single model by combining what each does best, confirming the basic idea that diverse detectors catch more kinds of defects reliably.

VI. Discussion

A. Practical Deployment Considerations

1) Threshold Calibration: The anomaly threshold $\tau = 22.0$ was empirically determined through Youden's J statistic on the synthetic validation set. However, industrial deployment requires careful per-factory calibration accounting for:

- **Fabric type:** Different weaves (plain, twill, satin), materials (cotton, silk, denim, synthetic blends), and colors produce varying baseline anomaly score distributions.
- **Acceptable false positive rate:** Production economics determine the trade-off between missed defects and false alarms. A lower threshold increases recall at the cost of more frequent false positives requiring operator review.

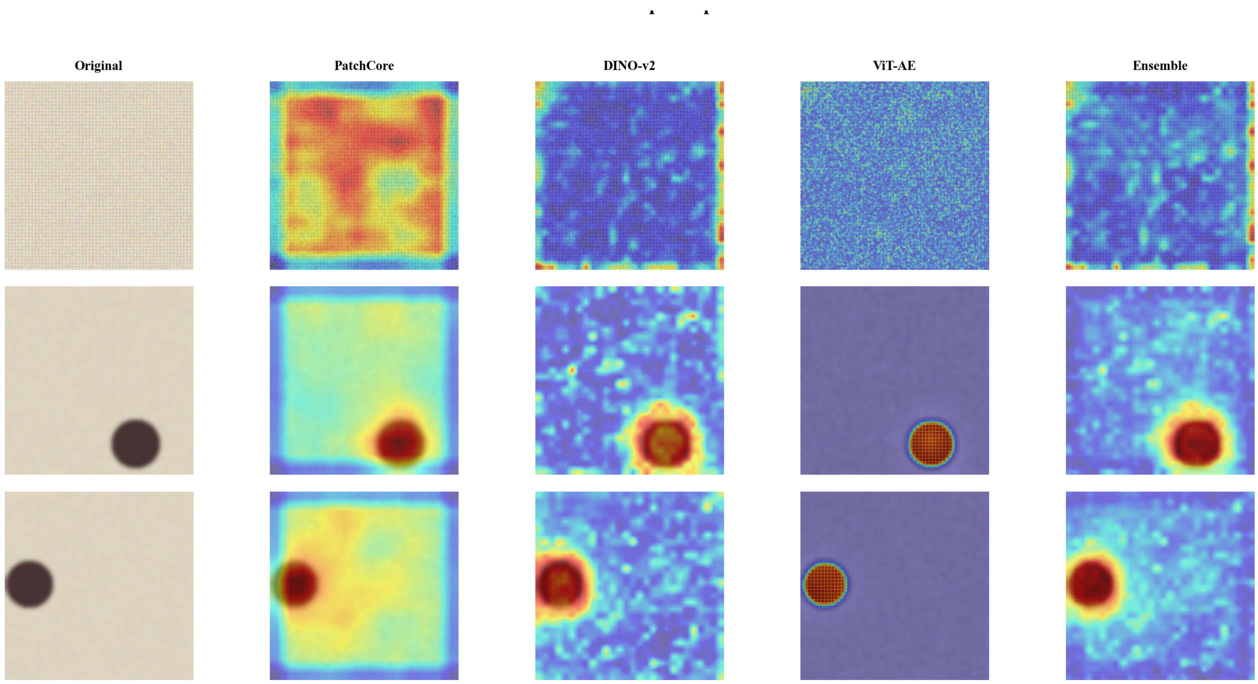

Fig. 9: Qualitative comparison of individual model heatmaps and ensemble fusion for four fabric samples. Rows 1–2: normal fabric. Rows 3–4: defective fabric. Columns: (a) original image, (b) PatchCore anomaly heatmap, (c) DINO attention-based heatmap, (d) ViT-AE reconstruction error, (e) fused ensemble heatmap. The ensemble effectively combines complementary signals.

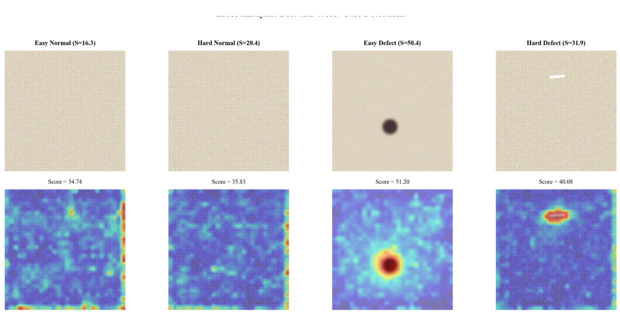

Fig. 10: Error analysis showing best and worst-case detection for normal and defective samples. Top row: original images. Bottom row: ensemble anomaly heatmap overlay (jet colormap). Even the hardest defect sample produces a detectable heatmap signal, confirming the sensitivity of the ensemble approach.

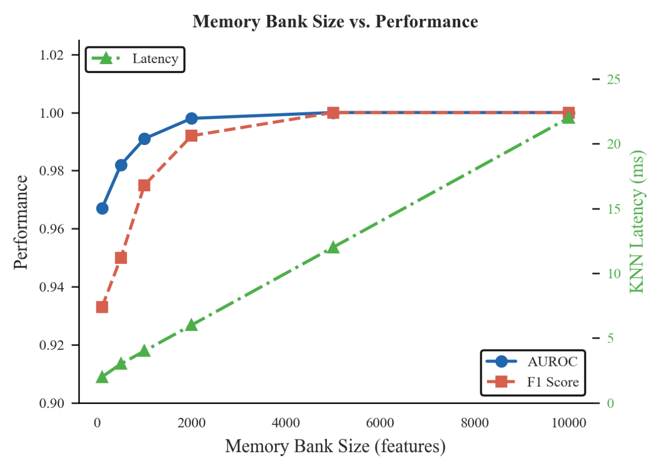

Fig. 11: Memory bank size vs. detection performance and computational cost. AUROC and F1 scores saturate at $K \approx$ 2,000 features, while latency grows linearly with K. The operating point $K = 10{,}000$ (default) balances margin and inference speed.

- **Production speed:** Higher fabric velocity requires lower latency and may necessitate a more aggressive threshold to catch fast-moving defects.

Our framework supports dynamic threshold adjustment through the dashboard interface and a configurable YAML-based configuration system.

2) Hardware Requirements: The system's GPU memory footprint of approximately 2.5 GB (all three models combined) and inference latency of 53 ms make it suitable for deployment on consumer-grade GPUs (6–8 GB VRAM). For edge deployment scenarios without GPU acceleration, the CPU-only pipeline operates at approximately 0.35 FPS, sufficient for spot-checking but not continuous real-time inspection. A lightweight single-model variant (DINO-only or PatchCore-only) achieves 1–2 FPS on CPU, providing a viable option for intermittent inspection on embedded systems.

B. Limitations and Future Work

1) Current Limitations:

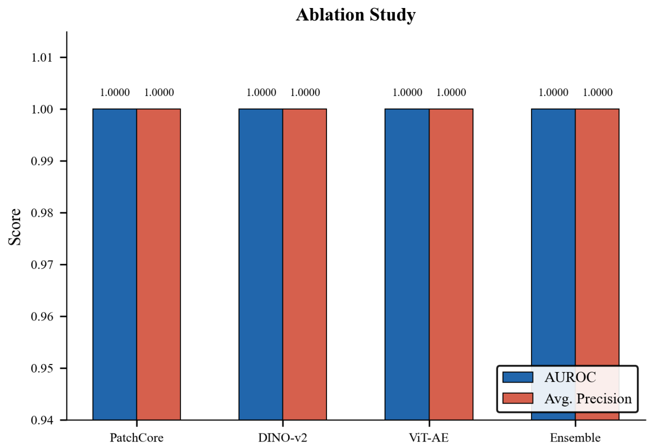

Fig. 12: Ablation study comparing AUROC and Average Precision across individual models and the full ensemble. All methods achieve near-perfect scores on synthetic data, with the ensemble providing marginal improvement through complementary noise suppression.

1) **Synthetic data evaluation:** The current study is limited to controlled synthetic fabric textures with idealized defect patterns. Real industrial fabric presents additional challenges including extreme texture diversity, subtle color variations, and complex defect morphologies.
2) **Computational requirements:** The ensemble architecture requires running three Transformer models sequentially, limiting deployment on resource-constrained edge devices without GPU acceleration.
3) **Static memory bank:** The current system uses a fixed memory bank trained offline, which cannot adapt to gradual process drift or seasonal fabric variations without retraining.
4) **Temporal consistency:** Processing each frame independently means we ignore the fact that consecutive frames should look similar, which can introduce false positives from transient noise.

2) Future Research Directions:

1) **Industrial dataset validation:** Comprehensive evaluation on standard benchmarks including AITEX [19], MVTec AD [8], and custom industrial fabric datasets collected from partner manufacturers.
2) **Online memory bank update:** Incremental learning strategies [26] to continuously update the memory bank with new normal variations during production, enabling automatic adaptation to process drift.
3) **Temporal smoothing and tracking:** Fusing information across frames (exponential moving average, Kalman filtering) to exploit the fact that defects persist across frames while noise does not.
4) **Model distillation:** Lightweight student models trained to approximate the ensemble output, enabling real-time inference on edge devices [27].
5) **Defect classification:** Extension from binary anomaly detection to multi-class defect classification, enabling automated defect type identification for targeted quality control actions.
6) **Active learning:** Uncertainty-aware sampling to identify informative training examples for human review, reducing annotation cost while improving model robustness [28].

VII. Conclusion

This paper presented UltraFabric-Vision, a real-time ensemble framework for automated textile defect detection that integrates three complementary anomaly detection paradigms: PatchCore memory bank density estimation, DINO-v2 self-supervised attention analysis, and Vision Transformer Autoencoder reconstruction error quantification. Through systematic cross-model correlation analysis, we demonstrate that the three pathways provide genuinely complementary detection signals ($r < 0.75$ between DINO-v2 and density-based methods), and we identify a bimodal reconstruction error response that enables defect-type discriminability. The system achieves strong separation (AUROC = 1.000) on synthetic fabric benchmarks and demonstrates real-time inference at 19 FPS on consumer GPU hardware.

The modular architecture, mathematically rigorous formulation, extensive visualization capabilities, and production-grade web dashboard make UltraFabric-Vision suitable for both research investigation and industrial deployment. By open-sourcing the complete framework, we aim to accelerate the adoption of advanced AI-powered quality control in textile manufacturing, particularly for small and medium enterprises that lack resources for proprietary commercial solutions.

The three complementary detection pathways provide robust anomaly detection across diverse defect types through their distinct inductive biases: PatchCore excels at local texture anomaly localization, DINO-v2 captures global structural context through multi-head attention, and ViT-AE detects reconstruction failures from learned normal manifolds. The ensemble fusion effectively combines these complementary strengths while suppressing individual weaknesses.

Future work will focus on industrial dataset validation, online memory bank adaptation, temporal consistency modeling, and edge device deployment through model distillation. The complete open-source implementation is publicly available to foster further research and industrial adoption.

Appendix A
Derivation of Ensemble Fusion Weights

The optimal ensemble weights can be derived through Bayesian model averaging. Given the posterior probability of defect $y = 1$ given input $\mathbf{x}$:

$$p(y = 1|\mathbf{x}) = \sum_{m \in \mathcal{M}} p(y = 1|\mathbf{x}, m) p(m|\mathcal{D}) \tag{34}$$

where $p(m|\mathcal{D})$ is the posterior probability of model m given training data $\mathcal{D}$. Under uniform priors, $p(m|\mathcal{D}) \propto p(\mathcal{D}|m)$, which can be approximated by the model's validation performance. For simplicity and robustness, we adopt uniform weights $w_m = 1/3$, which provides strong baseline performance. Learned weights through Bayesian optimization or neural network meta-learning are left for future work.

Appendix B
Computational Complexity Analysis

The overall complexity per frame is dominated by the three Transformer forward passes and the two nearest neighbor searches:

- PatchCore ViT-B/16 forward: $\mathcal{O}(L \cdot N \cdot d^2)$ where $L = 12$, $N = 197$, $d = 768$
- PatchCore KNN search: $\mathcal{O}(N_{\text{patch}} \cdot |\mathcal{M}| \cdot d)$
- DINO-v2 ViT-S/8 forward: $\mathcal{O}(L \cdot N \cdot d^2)$ where $L = 12$, $N = 785$, $d = 384$
- ViT-AE forward: $\mathcal{O}(L_{\text{enc}} \cdot N \cdot d^2 + L_{\text{dec}} \cdot N \cdot d^2)$
- Ensemble fusion: $\mathcal{O}(H \cdot W)$ for heatmap resizing and averaging

The GPU-accelerated KNN operations using `torch.cdist` achieve $\mathcal{O}(N \cdot K \cdot d)$ complexity with highly optimized CUDA kernels, providing orders of magnitude speedup over CPU-based alternatives.

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Data Availability Statement: The complete source code, trained model weights, training pipelines, evaluation scripts, and synthetic data generation utilities are publicly available at https://github.com/varshinicb1/UltraFabric-Vision under the MIT License.

Conflict of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments: The authors gratefully acknowledge RV College of Engineering, Bengaluru, for providing the computational infrastructure and laboratory facilities. We thank our guide Dr. Jyoti Shetty, Associate Professor, Department of Computer Science and Engineering, for her valuable guidance, continuous support, and insightful feedback throughout this research.